MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 16: Complex Series: Examples and Singularities

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Lecture outline

1. Singularities
2. Examples

Laurent Series Theorem - Review

Theorem

Laurent Expansion *Let $0 \leq r < R \leq \infty$ and suppose f is differentiable the annulus $r < |z - z_0| < R$. Then for each z in this annulus,*

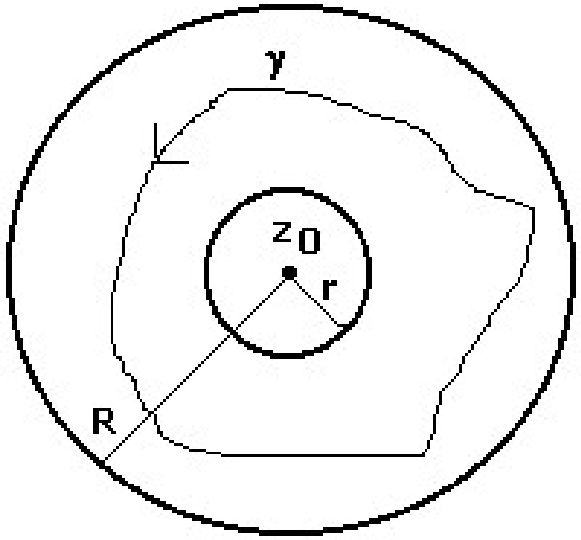
$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - z_0)^n \tag{1}$$

where for each integer n

$$c_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz \tag{2}$$

where γ is any closed path about z_0 lying entirely in the annulus.

Laurent Series Theorem - Review. Part 2



Remark: Just as in the case of Taylor series, one rarely finds the Laurent expansion of a function using the expression given by (2) . In fact, we shall actually see later that one can use one of these coefficients to evaluate integrals. Usually, one uses a known series to generate other series. This can be done since the next theorem assures us that the Laurent expansion of a function about a point is unique.

Theorem uniqueness

Theorem

Suppose f is differentiable in an annulus $r < |z - z_0| < R$, and for z in this annulus

$$f(z) = \sum_{n=-\infty}^{\infty} b_n (z - z_0)^n .$$

Then the b_n's are the Laurent coefficients of f about z_0.

Example 1

The function $e^{\frac{1}{z}}$ is differentiable in the annulus $0 < |z| < \infty$. Now

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

$$\implies \qquad e^{\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{1}{z^n} = 1 + \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{6z^3} + \cdots$$

This is the Laurent expansion of $e^{\frac{1}{z}}$ about 0, and it converges for $0 < |z| < \infty$.

Example 2

Find the Laurent expansion of $\dfrac{1}{1+z^2}$ about $z=-i$.

We want an expansion in powers of $z+i$. Note that the function is differentiable in the annulus $0<|z+i|<2$, where 2 is the distance from $-i$ to i, the other point at which f is singular. To get powers of $z+i$ we use partial fractions to get

$$\frac{1}{1+z^2}=\frac{i}{2}\frac{1}{z+i}-\frac{i}{2}\frac{1}{z-i}$$

The first term in the partial fractions expansion is a power of $z+i$ so we leave it. Also,

$$\frac{1}{z-i}=\frac{1}{-2i+z+i}=\frac{1}{-2i}\frac{1}{1-(z+i)/(2i)}=\frac{-1}{2i}\sum_{n=0}^{\infty}\left(\frac{z+i}{2i}\right)^n=-\sum_{n=0}^{\infty}\frac{1}{(2i)^{n+1}}$$

provided $|(z+i)/2i|<1$, that is, provided $|z+i|<2$. Hence

$$\frac{1}{1+z^2}=\frac{i}{2}\frac{1}{z+i}+\frac{i}{2}\sum_{n=0}^{\infty}\frac{1}{(2i)^{n+1}}(z+i)^n=\frac{i}{2}\sum_{n=-1}^{\infty}\frac{1}{(2i)^{n+1}}(z+i)^n$$

This is the Laurent expansion of $1/\left(1+z^2\right)$ in $0<|z+i|<2$.

Example 3

Show

$$\frac{1}{(1-z)(2-z)} = \sum_{n=2}^{\infty}\left(\frac{2^{n-1}-1}{z^n}\right) \qquad (2 < |z| < \infty).$$

$$\frac{1}{(1-z)(2-z)} = \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{z\left(1-\frac{2}{z}\right)} - \frac{1}{z\left(1-\frac{1}{z}\right)}$$

$$= \frac{1}{z}\left(\sum_{m=0}^{\infty}\frac{2^m}{z^m} - \sum_{m=0}^{\infty}\frac{1}{z^m}\right) = \frac{1}{z}\left(\sum_{m=1}^{\infty}\frac{2^m}{z^m} - \sum_{m=1}^{\infty}\frac{1}{z^m}\right)$$

$$= \sum_{m=1}^{\infty}\left(\frac{2^m-1}{z^{m+1}}\right) = \sum_{n=2}^{\infty}\left(\frac{2^{n-1}-1}{z^n}\right) \; n = m+1$$

Example 4.

Show that

$$\frac{1}{(1-z)(2-z)} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} - \sum_{n=1}^{\infty} \frac{1}{z^n}$$

for $1 < |z| < 2$

$$\frac{1}{(1-z)(2-z)} = \frac{1}{(z-1)(z-2)} = \frac{-1}{2-z} - \frac{1}{z-1} = \frac{-1}{2}\frac{1}{1-\frac{z}{2}} - \frac{1}{z}\frac{1}{1-\frac{1}{z}} =$$

$$= \frac{-1}{2}\sum_{n=0}^{\infty} \frac{z^n}{2^n} - \frac{1}{z}\sum_{n=0}^{\infty} \frac{1}{z^n} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} - \sum_{n=1}^{\infty} \frac{1}{z^n}$$

Example 5. Part 1

Find the Laurent expansion of the function

$$f(z) = \frac{7z-2}{z(z+1)(z-2)}$$

We decompose the fraction:

$$f(z) = \frac{7z-2}{z(z+1)(z-2)} =$$

$$\frac{1}{z} - \frac{3}{z+1} + \frac{2}{-2+z} = \frac{-1}{1-(z+1)} - \frac{3}{z+1} + \frac{2}{-3+(z+1)}$$

Remark: The term $\frac{-3}{z+1}$ is the Laurent expansion of itself about -1.

Example 5. Part 2

(a) in the annulus $1 < |z+1| < 3$.

$$\frac{-1}{1-(z+1)} - \frac{3}{z+1} + \frac{2}{-3+(z+1)} =$$

$$= \frac{-3}{z+1} + \frac{-1}{z+1}\frac{1}{\frac{1}{z+1}-1} + \frac{2}{3}\frac{1}{-1+\frac{z+1}{3}}$$

$$= \frac{-3}{z+1} + \frac{1}{z+1}\frac{1}{1-\frac{1}{z+1}} - \frac{2}{3}\frac{1}{1-\frac{z+1}{3}}$$

$$= \frac{-3}{z+1} + \frac{1}{z+1}\sum_{n=0}^{\infty}\frac{1}{(z+1)^n} - \frac{2}{3}\sum_{n=0}^{\infty}\frac{(z+1)^n}{3^n}$$

$$= \frac{-3}{z+1} + \sum_{n=0}^{\infty}\frac{1}{(z+1)^{n+1}} - 2\sum_{n=0}^{\infty}\frac{(z+1)^n}{3^{n+1}}$$

Example 5. Part 3

(b) in the annulus $0 < |z+1| < 1$

$$\frac{-1}{1-(z+1)} - \frac{3}{z+1} + \frac{2}{-3+(z+1)} =$$

$$= \frac{-3}{z+1} + \frac{-1}{1-(z+1)} + \frac{2}{-3}\frac{1}{1-\frac{(z+1)}{3}}$$

$$= \frac{-3}{z+1} - \sum_{n=0}^{\infty}(z+1)^n + \frac{-2}{3}\sum_{n=0}^{\infty}\frac{(z+1)^n}{3^n} =$$

$$= \frac{-3}{z+1} - \sum_{n=0}^{\infty}(z+1)^n - 2\sum_{n=0}^{\infty}\frac{(z+1)^n}{3^{n+1}}$$

Example 5. Part 4

(c) for $|z+1| > 3$

$$\frac{-1}{1-(z+1)} - \frac{3}{z+1} + \frac{2}{-3+(z+1)} = \frac{-3}{z+1} + \frac{-1}{z+1}\frac{1}{\frac{1}{z+1}-1} + \frac{2}{z+1}\frac{1}{1-\frac{3}{z+1}}$$

$$= \frac{-3}{z+1} + \frac{-1}{z+1}\frac{1}{\frac{1}{z+1}-1} + \frac{2}{z+1}\frac{1}{1-\frac{3}{z+1}} =$$

$$\frac{-3}{z+1} + \frac{1}{z+1}\frac{1}{1-\frac{1}{z+1}} + \frac{2}{z+1}\frac{1}{1-\frac{3}{z+1}}$$

$$= \frac{-3}{z+1} + \frac{1}{z+1}\sum_{n=0}^{\infty}\frac{1}{(z+1)^n} + \frac{2}{z+1}\sum_{n=0}^{\infty}\frac{3^n}{(z+1)^n}$$

$$= \frac{-3}{z+1} + \sum_{n=0}^{\infty}\frac{1}{(z+1)^{n+1}} + 2\sum_{n=0}^{\infty}\frac{3^n}{(z+1)^{n+1}}$$

Example 6.

Use the expansion from Example 2 (a) to show that

$$\oint_C \frac{7z-2}{z\,(z+1)\,(z-2)}dz = -4\pi i$$

where C is any closed curve lying in the interior of the circle $|z+1| = 3$ and enclosing the circle $|z+1| = 1$. (Hint: Consider the coefficient of $(z+1)^{-1}$ in the Laurent expansion of the integrand.)

Remark:

$$c_{-1} = \frac{1}{2\pi i}\int_C f(z)dz$$

from 2a)

$$c_{-1} = -3+1 = -2 \Rightarrow \int_C f(z)dz = -2(2\pi i) = -4\pi i$$

Example 7.

Show that $f(z) = \dfrac{1-\cos z}{z}$ for $z \neq 0$ has a removable singularity at 0.
Consider $\cos z$:

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!} \Rightarrow 1 - \cos z = 1 - \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!} \Rightarrow$$

$$\Rightarrow \frac{1-\cos z}{z} = \frac{1}{z} - \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n-1}}{(2n)!} =$$

$$= \frac{1}{z} - (\frac{1}{z} - \frac{z}{2!} + \frac{z^3}{4!} + \ldots)$$

It follows that $c_n = 0$ for $n < -2$. If $n = -1$, we have $\dfrac{1}{z} - \dfrac{1}{z} = 0$.
If we define $f(0) = 0$ then we remove the singularity and extend f to a function that is differentiable at 0.

Example 8.

Show that

(a) $f(z) = \dfrac{\sin z}{z^3}$ has a double pole at 0.

$$\frac{\sin z}{z^3} = \frac{1}{z^2}\frac{\sin z}{z}$$

From the notes $\dfrac{\sin z}{z}$ has a removable singularity at 0. But $\dfrac{1}{z^2}$ has a double pole at 0. So therefore $\dfrac{\sin z}{z^3}$ has a double pole at 0.

(b) $f(z) = e^{\frac{1}{z}}$ has an essential singularity at 0.

$$e^{\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{n!}\frac{1}{z^n} = 1 + \frac{1}{z} + \frac{1}{2z^2} + \ldots$$

One can see easily that $c_{-n} \neq 0$ for $n > 0$. Done.

Thank you